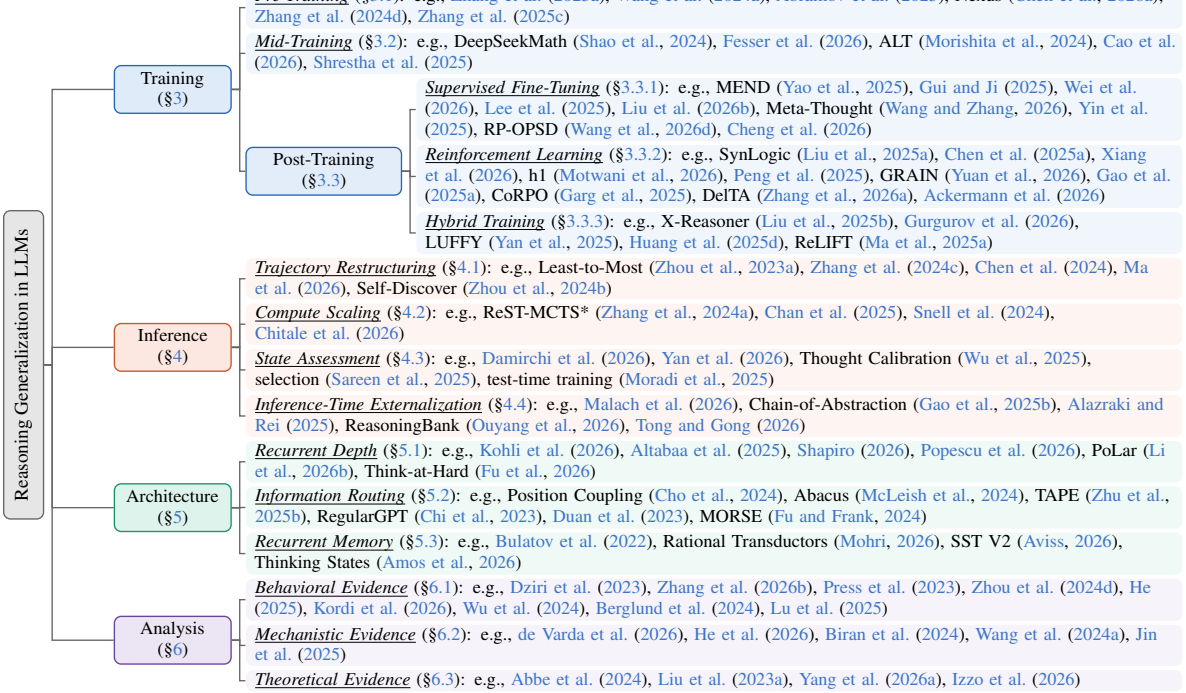

Reasoning Generalization in LLMs
Training (§3)
Pre-Training (§3.1): e.g., Zhang et al. (2025a), Wang et al. (2024a), Abramov et al. (2025), Nexus (Chen et al., 2026a), Zhang et al. (2024d), Zhang et al. (2025c)
Mid-Training (§3.2): e.g., DeepSeekMath (Shao et al., 2024), Fesser et al. (2026), ALT (Morishita et al., 2024), Cao et al. (2026), Shrestha et al. (2025)
Post-Training (§3.3)
Supervised Fine-Tuning (§3.3.1): e.g., MEND (Yao et al., 2025), Gui and Ji (2025), Wei et al. (2026), Lee et al. (2025), Liu et al. (2026b), Meta-Thought (Wang and Zhang, 2026), Yin et al. (2025), RP-OPSD (Wang et al., 2026d), Cheng et al. (2026)
Reinforcement Learning (§3.3.2): e.g., SynLogic (Liu et al., 2025a), Chen et al. (2025a), Xiang et al. (2026), h1 (Motwani et al., 2026), Peng et al. (2025), GRAIN (Yuan et al., 2026), Gao et al. (2025a), CoRPO (Garg et al., 2025), DelTA (Zhang et al., 2026a), Ackermann et al. (2026)
Hybrid Training (§3.3.3): e.g., X-Reasoner (Liu et al., 2025b), Gurgurov et al. (2026), LUFFY (Yan et al., 2025), Huang et al. (2025d), ReLIFT (Ma et al., 2025a)
Inference (§4)
Trajectory Restructuring (§4.1): e.g., Least-to-Most (Zhou et al., 2023a), Zhang et al. (2024c), Chen et al. (2024), Ma et al. (2026), Self-Discover (Zhou et al., 2024b)
Compute Scaling (§4.2): e.g., ReST-MCTS* (Zhang et al., 2024a), Chan et al. (2025), Snell et al. (2024), Chitale et al. (2026)
State Assessment (§4.3): e.g., Damirchi et al. (2026), Yan et al. (2026), Thought Calibration (Wu et al., 2025), selection (Sareen et al., 2025), test-time training (Moradi et al., 2025)
Inference-Time Externalization (§4.4): e.g., Malach et al. (2026), Chain-of-Abstraction (Gao et al., 2025b), Alazraki and Rei (2025), ReasoningBank (Ouyang et al., 2026), Tong and Gong (2026)
Architecture (§5)
Recurrent Depth (§5.1): e.g., Kohli et al. (2026), Altabaa et al. (2025), Shapiro (2026), Popescu et al. (2026), PoLar (Li et al., 2026b), Think-at-Hard (Fu et al., 2026)
Information Routing (§5.2): e.g., Position Coupling (Cho et al., 2024), Abacus (McLeish et al., 2024), TAPE (Zhu et al., 2025b), RegularGPT (Chi et al., 2023), Duan et al. (2023), MORSE (Fu and Frank, 2024)
Recurrent Memory (§5.3): e.g., Bulatov et al. (2022), Rational Transductors (Mohri, 2026), SST V2 (Aviss, 2026), Thinking States (Amos et al., 2026)
Analysis (§6)
Behavioral Evidence (§6.1): e.g., Dziri et al. (2023), Zhang et al. (2026b), Press et al. (2023), Zhou et al. (2024d), He (2025), Kordi et al. (2026), Wu et al. (2024), Berglund et al. (2024), Lu et al. (2025)
Mechanistic Evidence (§6.2): e.g., de Varda et al. (2026), He et al. (2026), Biran et al. (2024), Wang et al. (2024a), Jin et al. (2025)
Theoretical Evidence (§6.3): e.g., Abbe et al. (2024), Liu et al. (2023a), Yang et al. (2026a), Izzo et al. (2026)